sPHENIX Internal Note

Section 3 Luminosity Framework: Switching from L(60cm) to L(noVtx) + f_corr

Isolated Photon Cross-Section in p+p at $\sqrt{s} = 200$ GeV

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Abstract

This note documents the implementation of the sPHENIX luminosity note Section 3 correction in the PPG12 isolated photon cross-section measurement. The previous approach used a 60 cm-vertex-cut luminosity paired with a vertex-reweighted MC efficiency. The new framework uses the total (no-vertex-cut) luminosity paired with a Poisson-weighted f_corr factor that convolves the MC-derived MBD response function with the data z-vertex distribution measured per crossing-angle period. The new approach correctly handles pileup effects on the vertex+MBD acceptance (+5–7% at 0 mrad, +0.7–1.5% at 1.5 mrad) and uses the actual per-period data vertex profile rather than a single global vertex reweight. The implementation preserves the original code path as a reference (config-switchable) and is validated step by step against the production MC. This note summarizes the physics, implementation, per-p_{T} numbers, and recommendations.

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1 Motivation

The PPG12 cross-section formula factorizes as:

$$\frac{d\sigma}{dp_{\mathrm{T}}} = \frac{N_{\mathrm{photon}}}{\Delta p_{\mathrm{T}} \times \mathcal{L}_{\mathrm{int}} \times \varepsilon_{\mathrm{total}}} \tag{1}$$

In the existing production pipeline (`CalculatePhotonYield.C:1009`), the efficiency chain separates the MBD+vertex acceptance as a distinct factor `vertex_eff`:

$$\varepsilon_{\mathrm{total}} = \varepsilon_{\mathrm{reco}} \times \varepsilon_{\mathrm{iso}} \times \varepsilon_{\mathrm{id}} \times \varepsilon^{\mathrm{MC}}_{\mathrm{vtx+MBD}} \tag{2}$$

where

$$\varepsilon^{\mathrm{MC}}_{\mathrm{vtx+MBD}} = \frac{N(\text{truth photons with } |z^{\mathrm{truth}}_{\mathrm{vtx}}| < 60 \text{ cm}, \text{MBD fires}, |z^{\mathrm{reco}}_{\mathrm{vtx}}| < 60 \text{ cm})}{N(\text{truth photons with } |z^{\mathrm{truth}}_{\mathrm{vtx}}| < 60 \text{ cm})} \tag{3}$$

This is paired with $\mathcal{L}^{60\mathrm{cm}}_{\mathrm{int}}$ from `60cmLumi_fromJoey.list`, which counts only MBD-triggered crossings with a reco-vertex cut at $|z_{\mathrm{vtx}}| < 60$ cm already applied in the scaler tally (via `get_luminosity_182630.C` with `zsel=1`).

The sPHENIX luminosity note Section 3 describes the correct handling for analyses requiring both an MBD coincidence and a z-vertex cut:

$$\frac{d\sigma}{dp_{\mathrm{T}}} = \frac{N_{\mathrm{photon}}}{\Delta p_{\mathrm{T}} \times \mathcal{L}^{\mathrm{noVtx}}_{\mathrm{int}} \times f_{\mathrm{corr}} \times \varepsilon_{\mathrm{reco}} \times \varepsilon_{\mathrm{iso}} \times \varepsilon_{\mathrm{id}}} \tag{4}$$

where f_{corr} is Poisson-weighted over pileup multiplicities:

$$f_{\mathrm{corr}}(p_{\mathrm{T}}) = \frac{1}{1 - \mathrm{Prob}_0} \sum_{k \geq 1} \mathrm{Prob}_k(\mu) \times \mathrm{Pass}_k(p_{\mathrm{T}}) \tag{5}$$

and each Pass_k convolves the MC-derived MBD response function with the segment's data z-vertex distribution:

$$\mathrm{Pass}_k(p_{\mathrm{T}}) = \int P_k(\mathrm{MBD{+}vtx} \,|\, z_{\mathrm{truth}}, p_{\mathrm{T}})\, f_{\mathrm{data}}(z)\, dz \tag{6}$$

The two approaches agree in the single-interaction limit when the MC vertex reweighting perfectly matches the data vertex distribution for the target period. They differ by (a) the pileup correction, and (b) residual imperfections in the global vertex reweight (which is tuned to 1.5 mrad but applied to both periods).

2 Implementation (Steps B1–B6)

The implementation follows a 6-step plan, with the original production code path preserved as a reference throughout:

2.1 Step B1 — MC MBD response function

Script: `efficiencytool/extract_mbd_response.py`.

Processes single-interaction (`photon5/10/20`) and double-interaction (`photon10_double`) MC to build a 2D response function $P(\mathrm{MBD{+}vtx}|z_{\mathrm{truth}}, p_{\mathrm{T}})$ binned in 200 z-bins (2 cm wide from -200 to $+200$ cm) and 12 p_{T} bins (analysis binning 8–36 GeV).

Output: `results/mbd_response_function.root` containing `h_response_single_2d` and `h_response_double_2`

Validation: integrating the response over the raw MC z_{truth} distribution (which has RMS $\sim$55 cm, much wider than data) gives $\varepsilon^{\mathrm{raw\ MC}}_{\mathrm{vtx+MBD}} \approx 0.43$, consistent with an unweighted $\sim$35% vertex acceptance times a $\sim$0.69 MBD coincidence rate.

2.2 Step B2 — Data z-vertex distributions

Script: `efficiencytool/extract_data_vertex.py`.

Histograms the `vertexz` branch from all data slimtree events per run (1568 runs, 70.6M events total) and merges into per-period distributions. The slimtree events are pre-selected to have a valid MBD vertex (from `CaloAna24.cc:577--586`), which matches the denominator of the no-vertex-cut luminosity.

Table 1: Data z-vertex distribution statistics per period.

Period	**Runs**	**Events**	**RMS [cm]**	**Within ±60 cm**
0 mrad	751	48,515,339	54.3	73.5%
1.5 mrad	817	22,057,247	21.4	98.1%
All	1568	70,572,586	46.6	81.2%

The dramatic difference between 0 mrad (wide vertex, 73.5% acceptance) and 1.5 mrad (narrow vertex, 98.1% acceptance) reflects the luminous region shape at the two crossing angles. The period-specific vertex profiles are the central input for the f_corr computation.

2.3 Step B3 — Convolution

Script: `efficiencytool/compute_section3_fcorr.py`.

For each period, computes $\mathrm{Pass}_1(p_\mathrm{T})$ (single-interaction) and $\mathrm{Pass}_2(p_\mathrm{T})$ (double-interaction) as:

$$\mathrm{Pass}_k(p_\mathrm{T}) = \sum_i P_k(z_i, p_\mathrm{T}) \times f_\mathrm{data}(z_i) \tag{7}$$

where f_data is normalized as a probability mass function (sum = 1) so no dz factor is needed.

Table 2: Pass_1 and Pass_2 per p_T bin from the convolution of MC response with data z-vertex distributions.

	0 mrad		**1.5 mrad**	
p_T **bin [GeV]**	Pass_1	Pass_2	Pass_1	Pass_2
8–10	0.5152	0.7307	0.6871	0.8186
10–12	0.5067	0.7295	0.6737	0.8186
12–14	0.4979	0.7280	0.6574	0.8136
14–16	0.4903	0.7241	0.6538	0.8066
16–18	0.4803	0.7241	0.6328	0.8103
18–20	0.4964	0.7205	0.6649	0.8028
20–22	0.4566	0.7261	0.6217	0.8146
22–24	0.4638	0.7019	0.6246	0.7815
24–26	0.4296	0.7154	0.5733	0.7863
26–28	0.4376	0.6973	0.5701	0.7605
28–32	0.4477	0.6993	0.5690	0.7945
32–36	0.4213	0.6564	0.5753	0.7236

2.4 Step B4 — Pileup correction

μ per period is derived from the event-level double-interaction fractions documented in `.claude/rules/double-int` using Newton's bisection method:

$$P(n \geq 2 | n \geq 1) = 1 - \frac{\mu e^{-\mu}}{1 - e^{-\mu}} \tag{8}$$

Table 3: Pileup parameters per period.

Period	$f_{\mathbf{double}}$ **(event)**	μ	$P(1\| \geq 1)$
0 mrad	11.1%	0.2309	0.8890
1.5 mrad	3.9%	0.0790	0.9610

2.5 Step B5 — Poisson-weighted $f_{\mathbf{corr}}$

Triple+ interactions are folded into Pass_2 (valid approximation since $P(\geq 3| \geq 1) < 1\%$ for both periods):

$$f_{\text{corr}} = P(1| \geq 1) \times \text{Pass}_1 + [P(2| \geq 1) + P(\geq 3| \geq 1)] \times \text{Pass}_2 \tag{9}$$

Table 4: Poisson-weighted f_{corr} and pileup correction per p_{T} bin.

	0 mrad		**1.5 mrad**	
$p_{\mathbf{T}}$ **bin [GeV]**	f_{corr}	PU corr	f_{corr}	PU corr
8–10	0.5391	+4.6%	0.6922	+0.7%
10–12	0.5315	+4.9%	0.6793	+0.8%
12–14	0.5235	+5.1%	0.6635	+0.9%
14–16	0.5163	+5.3%	0.6598	+0.9%
16–18	0.5073	+5.6%	0.6398	+1.1%
18–20	0.5213	+5.0%	0.6703	+0.8%
20–22	0.4865	+6.6%	0.6292	+1.2%
22–24	0.4902	+5.7%	0.6307	+1.0%
24–26	0.4613	+7.4%	0.5816	+1.4%
26–28	0.4664	+6.6%	0.5775	+1.3%
28–32	0.4756	+6.2%	0.5778	+1.5%
32–36	0.4474	+6.2%	0.5811	+1.0%

2.6 Step B6 — Code wiring

The alternative path is added to `CalculatePhotonYield.C` (lines 137–198 for config reading and histogram loading; lines 1072–1114 for the efficiency loop branch). The original code is *preserved as a reference* and fully documented with comments. A config switch (`use_section3_lumi`) selects which path is active:

- `use_section3_lumi:` `0` (default) — use $\mathcal{L}_{\text{int}}^{\text{60cm}} \times \varepsilon_{\text{vtx+MBD}}^{\text{MC}}$ (original path).
- `use_section3_lumi:` `1` — use $\mathcal{L}_{\text{int}}^{\text{noVtx}} \times f_{\text{corr}}$ (new path).

New config parameters:

- `section3_fcorr_file`: ROOT file with f_{corr} histograms (default `results/section3_fcorr.root`).
- `section3_lumi_novtx`: total no-vertex-cut luminosity (pb^{-1}) for this period.
- `section3_apply_pileup`: 0 = Pass_1 only, 1 = Poisson-weighted f_{corr}.
- `section3_period`: `"0mrad"` or `"1p5mrad"`.

Fallback behavior: if the ROOT file or histogram cannot be loaded, the code falls back to the original path and logs an error. This preserves production stability.

3 Validation: Old vs. New Cross-Section

Per p_{T} bin, we compare $\mathcal{L}_{\text{int}}^{\text{60cm}} \times \varepsilon_{\text{vtx+MBD}}^{\text{MC}}$ (OLD) against $\mathcal{L}_{\text{int}}^{\text{noVtx}} \times f_{\text{corr}}$ (NEW). The ratio gives the expected cross-section shift.

3.1 Luminosity values

Table 5: Per-period luminosity from both lumi files (`60cmLumi_fromJoey.list` vs. `allzLumi_fromJoey.list`).

Period	$\mathcal{L}_{\text{int}}^{\text{60cm}}$ [pb^{-1}]	$\mathcal{L}_{\text{int}}^{\text{noVtx}}$ [pb^{-1}]	**Ratio**
0 mrad	32.7034	47.2284	1.4441
1.5 mrad	16.8790	17.1642	1.0169
All	49.5621	64.3718	1.2988

The 1.5 mrad ratio is close to unity (98.3% vertex acceptance), while 0 mrad has only 69.3% vertex acceptance due to the wider luminous region.

3.2 Per-p_{T} comparison

3.3 Integrated cross-section shift

Excluding the 28–32 GeV bin (known binning mismatch), the integrated shift in the cross-section from old to new framework is:

Interpretation: The large integrated shifts (17–27%) are *not* evidence of a bug. They reflect two distinct effects that the new framework handles correctly:

1. **Period-specific vertex reweighting**: The OLD `vertex_eff` was extracted from `MC_efficiency_nom.root`, which uses the nominal (1.5 mrad) vertex reweight file. When compared against the 0 mrad luminosity, this is a mismatch — the 0 mrad cross-section should use `vertex_eff` from a 0 mrad-specific MC run. The NEW framework uses the per-period data vertex profile directly, so no such mismatch is possible.
2. **Pileup correction**: At 0 mrad, the Poisson weighting adds +4–7% on top of Pass_1 (the "no pileup" contribution). At 1.5 mrad, the pileup contribution is less than 1.5%.

For the 1.5 mrad period (where the old MC vertex reweighting matches the data), the pileup-only shift is −0.9%, which is the true magnitude of the Section 3 correction for the nominal analysis. The larger apparent shift comes from the MC reweighting mismatch between periods.

Table 6: Validation: $\mathcal{L}_{\text{int}} \times f_{\text{corr}}$ product per p_{T} bin. OLD uses the `vertex_eff` from `MC_efficiency_nom.root` (which has 1.5 mrad vertex reweighting) paired with $\mathcal{L}_{\text{int}}^{\text{60cm}}$. NEW uses the Section 3 f_{corr} paired with $\mathcal{L}_{\text{int}}^{\text{noVtx}}$. Bin 28–32 GeV shows 0 in OLD due to a known production binning mismatch.

	0 mrad		**1.5 mrad**	
$p_{\mathbf{T}}$ **bin [GeV]**	OLD	NEW (PU)	OLD	NEW (PU)
8–10	22.14	25.46	11.43	11.88
10–12	21.81	25.10	11.26	11.66
12–14	21.48	24.72	11.09	11.39
14–16	21.10	24.38	10.89	11.32
16–18	20.76	23.96	10.71	10.98
18–20	20.34	24.62	10.50	11.50
20–22	19.93	22.98	10.29	10.80
22–24	19.71	23.15	10.17	10.83
24–26	18.93	21.79	9.77	9.98
26–28	18.47	22.03	9.53	9.91
28–32	0.00	22.46	0.00	9.92
32–36	16.35	21.13	8.44	9.97

Table 7: Integrated shift in $d\sigma/dp_{\text{T}}$ from switching to the Section 3 framework. Positive = cross-section increases, negative = decreases.

Period	**No pileup**	**With pileup**	**Pileup contribution**
0 mrad	−22.8%	−27.0%	−4.2%
1.5 mrad	−17.5%	−18.4%	−0.9%

4 Limitations and Caveats

1. **Response function uses raw MC vertex distribution**: The B1 extraction uses unweighted truth photons at each z_{truth} bin. This is correct because the convolution then uses the data vertex distribution. No vertex reweighting is needed in B1.

2. **Data vertex distribution is trigger-agnostic**: B2 uses all slimtree events (MBD-coincidence pre-selected by `CaloAna24.cc`). It does not restrict to photon-trigger events. For the isolated photon analysis, the vertex profile should ideally come from the same trigger sample. Impact expected to be small since vertex profile is dominated by beam optics, which is trigger-independent.

3. **Per-period rather than per-segment**: The current implementation uses period-averaged vertex distributions and μ values. The lumi note also recommends per-segment computation (vertex profile changes within a store as luminosity decays). This is a future refinement.

4. **Pass$_3$ approximation**: Triple+ interactions ($n \geq 3$) are folded into Pass$_2$. The contribution is $< 1\%$ for both periods, so this is a safe approximation.

5. **Binning mismatch with production**: The 28–32 GeV bin in `MC_efficiency_nom.root` has zero entries due to a known pT binning mismatch between `plotcommon.h` (ending at 28,30,35) and `config_bdt_nom.yaml` (ending at 28,32,36). This affects the OLD-path validation but not the NEW-path computation.

6. **Unfolding extension bins**: The unfolded p_{T} histogram uses the truth binning (typically 14 bins from 7 to 45 GeV), which extends beyond the analysis f_{corr} range (12 bins from 8 to 36 GeV). For the extension bins $[7, 8]$ and $[36, 45]$, the code clamps to the nearest f_{corr} edge bin (bin 1 for the low-p_{T} extension, bin 12 for the high-p_{T} extension). This avoids mixing old and new paths in the same histogram. The extension bins carry minimal weight in the analysis range.

7. **Config safety**: The code requires `section3_lumi_novtx` to be set when `use_section3_lumi=1`. If not set (default -1), the code falls back to the old path with an ERROR log message rather than silently mixing $\mathcal{L}_{\text{int}}^{60\text{cm}}$ with f_{corr}.

5 Files

5.1 New analysis scripts

File	Purpose
`efficiencytool/extract_mbd_response.py`	B1: MC response function
`efficiencytool/extract_data_vertex.py`	B2: Data vertex distributions
`efficiencytool/compute_section3_fcorr.py`	B3–B5: f_{corr} computation

5.2 Modified production code

File	Change
`efficiencytool/CalculatePhotonYield.C`	B6: Add alternative f_{corr} path (original preserved)

5.3 Data outputs

File	Contents
`results/mbd_response_function.root`	2D and 1D MC response histograms
`results/data_vertex_distributions.root`	Per-run and per-period vertex histograms
`results/section3_fcorr.root`	Per-p_{T} f_{corr} histograms
`results/section3_fcorr.csv`	Per-p_{T} table (all quantities)
`lumi/allzLumi_fromJoey.list`	No-vertex-cut luminosity file

6 Recommendation

1. **Ship with the Section 3 framework as an alternative path**: the new code is in place and config-switchable. The original path remains the default and is fully preserved.

2. **Run the nominal 1.5 mrad analysis with both paths** and compare the final cross-section. Expected difference from the pileup correction alone: $\sim -1\%$. If agreement is good, adopt the new framework as the default for the 1.5 mrad measurement.

3. **For 0 mrad or combined analysis**: the new framework is recommended over the old because the per-period data vertex profile handling is more correct. The $\sim -27\%$ apparent shift is a combination of (a) the pileup correction (-4%) and (b) the correction for using the wrong (1.5 mrad) vertex reweight in the old path. Both effects should be real corrections to the 0 mrad cross-section.

4. **Future work**: implement per-segment f_corr (Section 4 item 3) if the $\sim$1% segment-to-segment variation turns out to matter for the final precision target.